ELEMENTA

CS23531 – WEB PROGRAMMING
MINI PROJECT REPORT

Submitted by

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BONAFIDE CERTIFICATE

Certified that this project report titled "**ELEMENTA – AN INTERACTIVE GUIDE TO THE ELEMENTS**" is the Bonafide work of **NARESH M (241001147)** who carried out the work under my supervision.

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INTERNAL EXAMINAR **EXTERNAL EXAMINAR**

ELEMENTA

ABSTRACT:

Elementa is an interactive periodic table of all 118 chemical elements built entirely using HTML5, CSS3 and Vanilla JavaScript, with no external libraries or frameworks. It presents every element in its correct IUPAC periodic table position using CSS Grid, where each element tile is placed dynamically from a JavaScript data array rather than hardcoded markup. Users can search elements in real time by name, symbol or atomic number, filter by chemical category using category pills, and click any tile to open a detailed specimen modal showing physical and atomic properties including electron configuration, melting point, boiling point, density, electronegativity and discovery history. The site supports full keyboard navigation, a persistent light/dark theme toggle backed by localStorage, and a random element picker. The design follows an editorial, archival aesthetic inspired by printed science books and museum catalogues, using the Fraunces, Plus Jakarta Sans and JetBrains Mono typefaces with an earthy, muted colour palette.

- Naresh M

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ELEMENTA

1.1 INTRODUCTION

The periodic table is one of the most fundamental tools in chemistry and science education. Most online periodic table resources are either outdated in design or too complex for students to navigate comfortably. Elementa is a web application that reimagines the periodic table as an interactive, editorial-style reference. All 118 elements are rendered in their correct IUPAC positions using CSS Grid, and their data is stored in a single JavaScript array rather than static HTML, making the codebase clean and maintainable. Users can search, filter by category and explore detailed property cards for any element, all without any page reloads.

1.2 PURPOSE

The purpose of Elementa is to provide a fast, accessible and visually clear periodic table reference for students and anyone studying chemistry. It aims to:

- Display all 118 elements in their exact standard periodic table layout using CSS Grid driven by dynamic JavaScript rendering.
- Allow users to search elements instantly by name, symbol or atomic number without reloading the page.
- Enable filtering by chemical category (alkali metals, noble gases, transition metals, etc.) with category pill toggles.
- Show detailed physical and atomic properties for each element in a clean modal overlay.
- Support full keyboard navigation (arrow keys to navigate elements, Escape to close modal) and screen reader accessibility.
- Persist the user's light/dark theme preference across sessions using localStorage.

1.3 SCOPE OF THE PROJECT

Elementa is a client-side, single-page web application intended for educational use. Its scope covers:

- **Element Data:** All 118 elements with atomic number, symbol, name, mass, category,

period, group, phase at STP, electron configuration, electronegativity, melting point, boiling point, density, discoverer, name etymology and a brief scientific summary.

- **Dynamic Rendering:** The periodic table grid and all element tiles are generated entirely by JavaScript at page load from the data array, with no hardcoded element divs in HTML.
- **Search:** Real-time filtering of all 118 element tiles by name, symbol or atomic number, with dimming of non-matching tiles and a live visible-count badge.
- **Category Filtering:** Ten category pill filters that can be combined with the search query through a single unified filter function.
- **Element Detail Modal:** A specimen card overlay for each element showing all stored properties, with previous/next navigation between elements.
- **Theme Toggle:** A light/dark mode switch with user preference stored in localStorage.
- **Responsive Layout:** The table scrolls horizontally on smaller screens to preserve scientific proportions.

The project does not include a backend, database, user accounts or server-side logic. It runs entirely in the browser from three static files.

1.4 SOFTWARE REQUIREMENTS OF THE PROJECT

Overall Description

Elementa runs entirely in the browser as a static web application. There is no server-side logic or database. The three files (index.html, style.css, script.js) can be opened directly in any modern browser or served from any static web host.

Product Perspective

The system is a single-tier, client-only application:

- **Presentation and Logic tier:** index.html defines the semantic structure; style.css defines all visual design, layout and theme tokens; script.js contains the full element dataset, DOM generation, event handling, search, filtering, modal control and theme management.

Hardware Requirements

- Processor: Any modern CPU
- RAM: 2 GB or higher
- Any device with a modern web browser

Software Requirements

- Browser: Google Chrome, Mozilla Firefox, Microsoft Edge or Safari (any modern version)
- No backend, server or database required
- No npm, build tools or frameworks required

Product Functionality

- **Render Table:** Generates all 118 element tiles in correct grid positions on page load.
- **Search:** Filters tiles in real time as the user types; highlights matches and dims non-matches.
- **Category Filter:** Toggles a category filter that combines with the active search query.
- **Element Modal:** Opens a detailed specimen card for any element with all physical and

atomic data.

- **Modal Navigation:** Previous/next buttons and left/right arrow keys walk through elements 1-118 inside the modal.
- **Random Picker:** Selects a random element and opens its detail modal.
- **Theme Toggle:** Switches between light and dark mode; preference is saved to localStorage.
- **Keyboard Accessibility:** Escape closes the modal; arrow keys navigate elements.

Assumptions and Dependencies

- A modern browser with ES6+ JavaScript support is available.
- Google Fonts (Fraunces, Plus Jakarta Sans, JetBrains Mono) are loaded from fonts.googleapis.com; if offline, system serif/sans-serif fallbacks are used.
- JavaScript must be enabled in the browser.
- No internet connection is required beyond loading Google Fonts on first visit.

Functional Requirements

Element Rendering Module (ERM)

- On page load, script.js iterates the 118-element data array and creates a button tile for each element.
- Each tile is placed at its exact CSS Grid position using the element's gridRow and gridCol properties.
- Lanthanide and Actinide series placeholders are injected into rows 6 and 7 of the main grid.

Search Module (SM)

- The search input fires on every keystroke.
- Each tile's data-name, data-symbol and data-number attributes are checked against the lowercased query.
- Matching tiles receive the .highlighted class; non-matching tiles receive .dimmed (opacity reduced to ~0.16).

- The visible-count badge updates on every search event.

Category Filter Module (CFM)

- Ten category pills are generated dynamically from unique category values in the data array.
- Clicking a pill sets the active category; clicking the active pill again toggles it off.
- The unified applyFiltersAndSearch() function evaluates both query and category simultaneously.

Modal Module (MM)

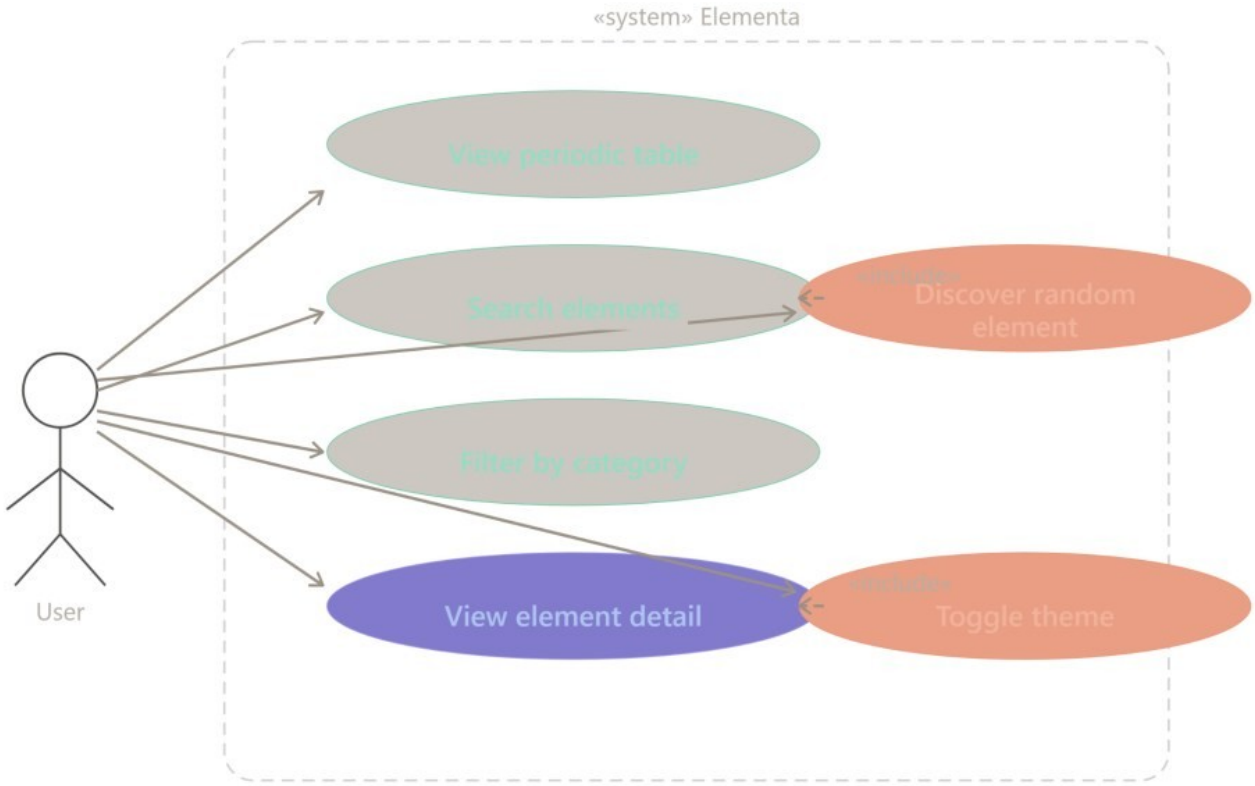
- Clicking any tile calls openElementDetail(number) which finds the element in the array by atomic number.
- All modal DOM fields are populated from the element object.
- Body scrolling is disabled while the modal is open.
- The modal closes on the close button click, backdrop click or Escape key.

Theme Module (TM)

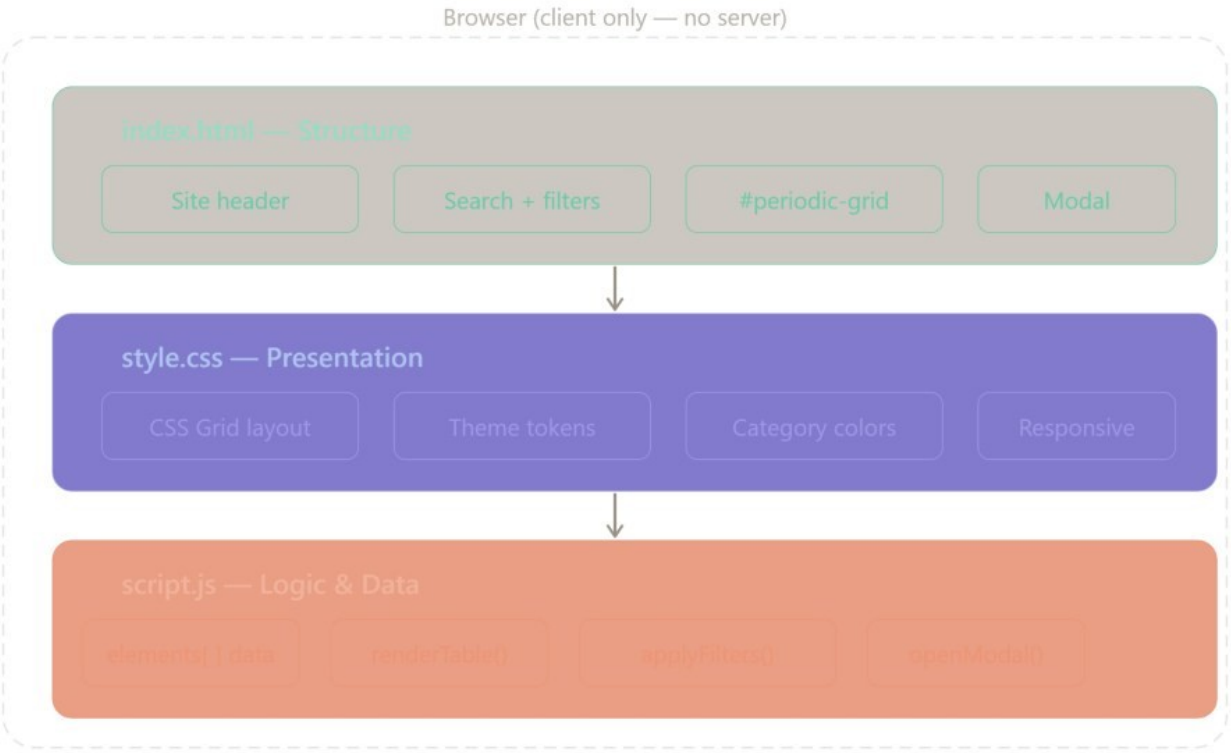
- On load, initTheme() reads localStorage.getItem('elementa-theme').
- The theme button toggles the data-theme attribute on the html element between light and dark.
- The chosen value is written back to localStorage on every toggle.

2. SYSTEM FLOW DIAGRAMS

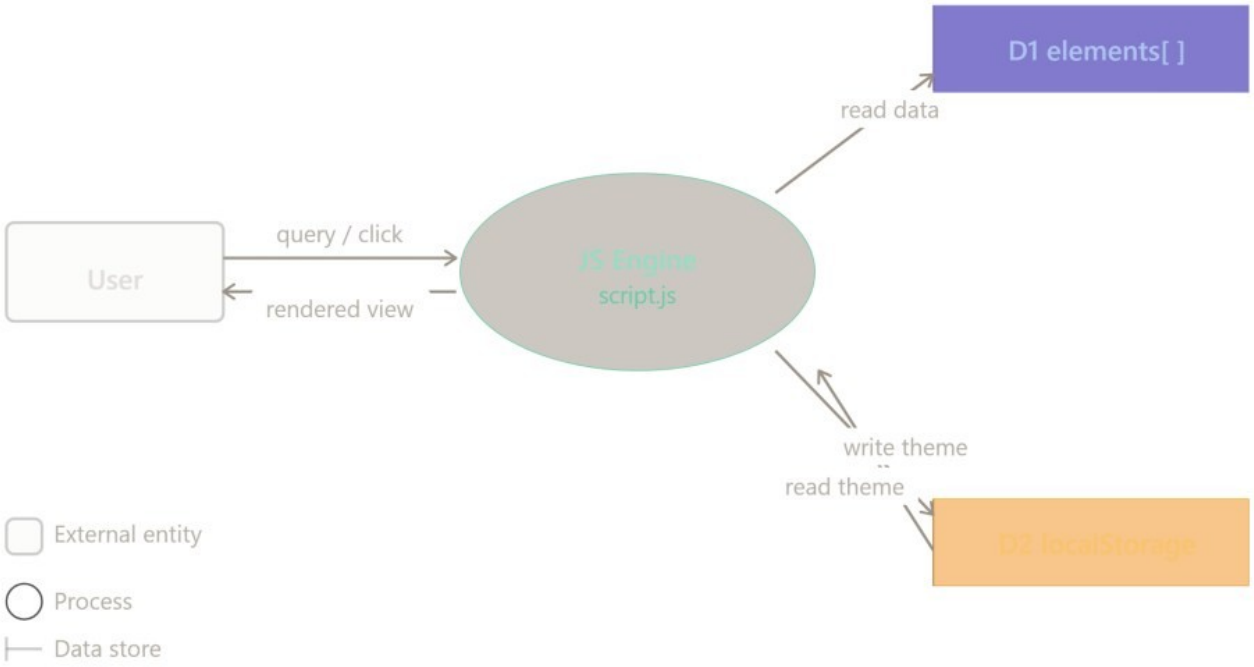
2.1 USE CASE DIAGRAM



2.2 SYSTEM ARCHITECTURE DIAGRAM



2.3 DATA FLOW DIAGRAM



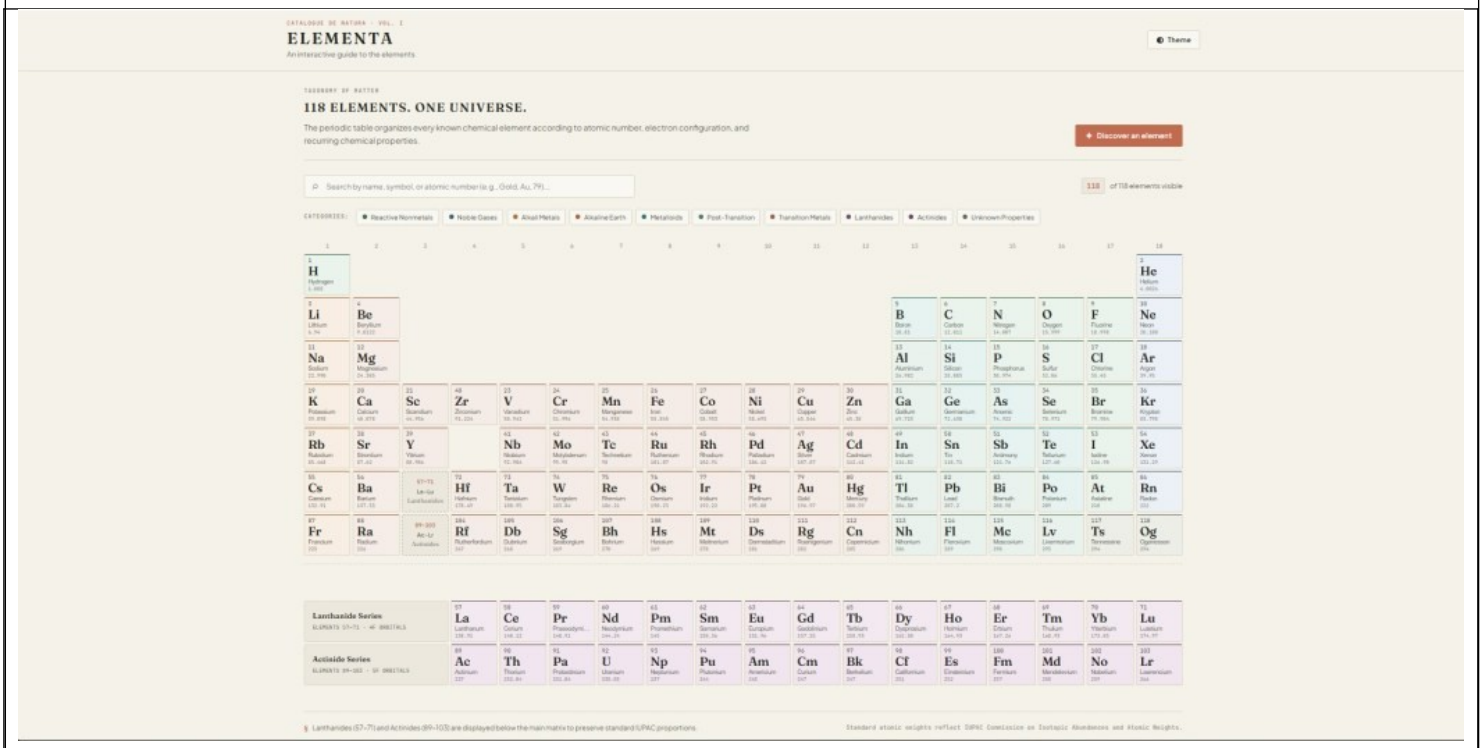
3. MODULE DESCRIPTION

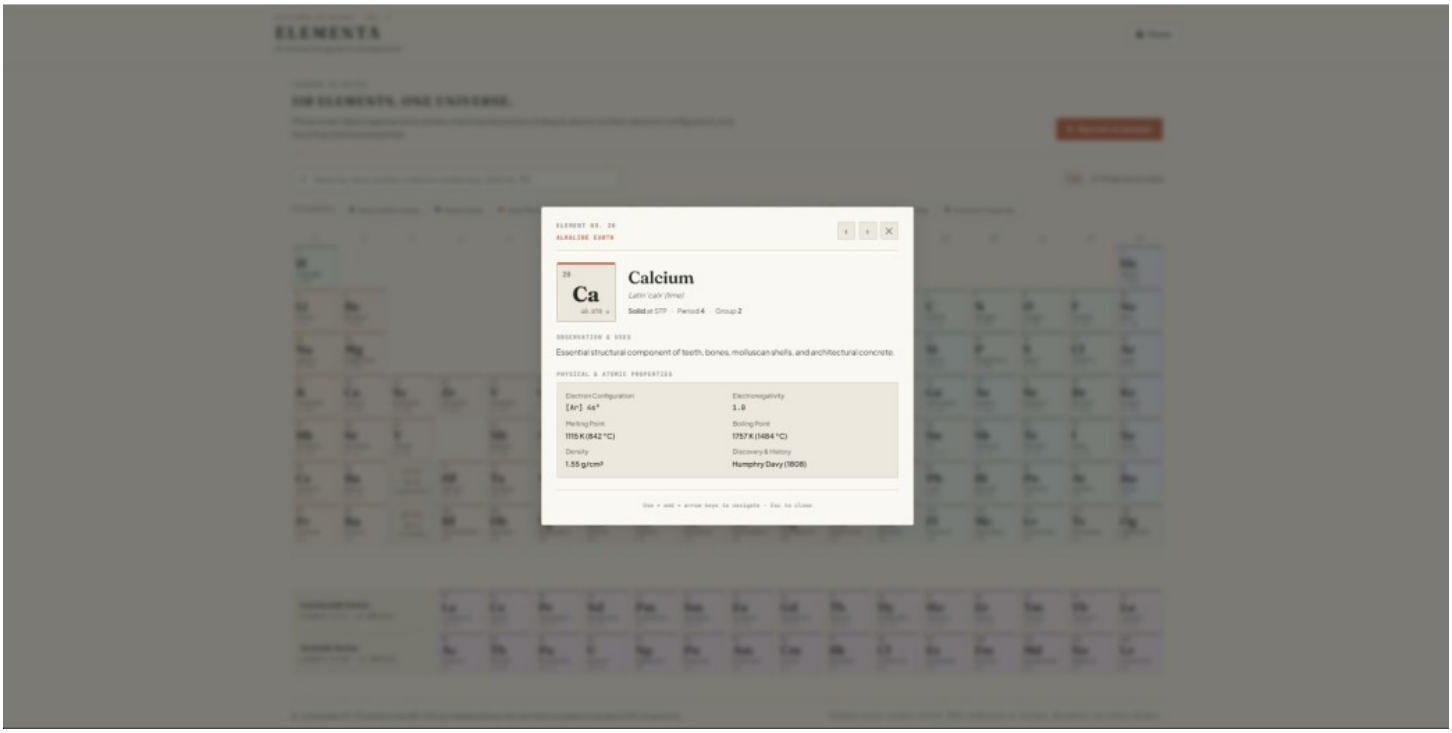
- **Element Data Array (script.js — const elements):** Stores the complete dataset of all 118 elements as an array of objects. Each object contains 15+ properties covering grid position, physical data, atomic data, history and description.
- **Rendering Module (renderPeriodicTable()):** Runs on page load. Clears the grid, injects Lanthanide/Actinide placeholder rows, loops the elements array, creates a button tile for each element, sets its CSS Grid position from gridRow/gridCol properties and attaches a click listener.
- **Filter and Search Module (applyFiltersAndSearch()):** Single unified function called by both the search input event and the category pill click event. Evaluates each tile against the active search query and the active category filter simultaneously, then shows or dims each tile and updates the count badge.
- **Modal Module (openElementDetail(), closeModal()):** Finds the element by atomic number, populates all modal fields, disables body scroll and shows the modal. Handles previous/next navigation and closes on Escape or backdrop click.
- **Category Pills Module:** Dynamically generates one pill button per unique category from the data array. Manages the active category state and routes to applyFiltersAndSearch() on click.
- **Theme Module (initTheme(), theme toggle button):** Reads and writes the theme preference to localStorage. Toggles the data-theme attribute on the html element to switch between light and dark CSS variable sets.
- **Random Picker:** Picks a random index from the elements array using Math.random() and calls openElementDetail() with that element's atomic number.
- **Stylesheet (style.css):** Defines all CSS custom properties (colour tokens, font stacks, spacing), the 18-column x 10-row CSS Grid layout, responsive horizontal scroll, element tile styles, category colour coding, modal styles and complete light/dark theme variable sets.
- **HTML Structure (index.html):** Semantic skeleton containing the site header, intro section, search/filter controls bar, table viewport container, legend footer and the detail modal markup. All element tiles are injected into #periodic-grid by script.js.

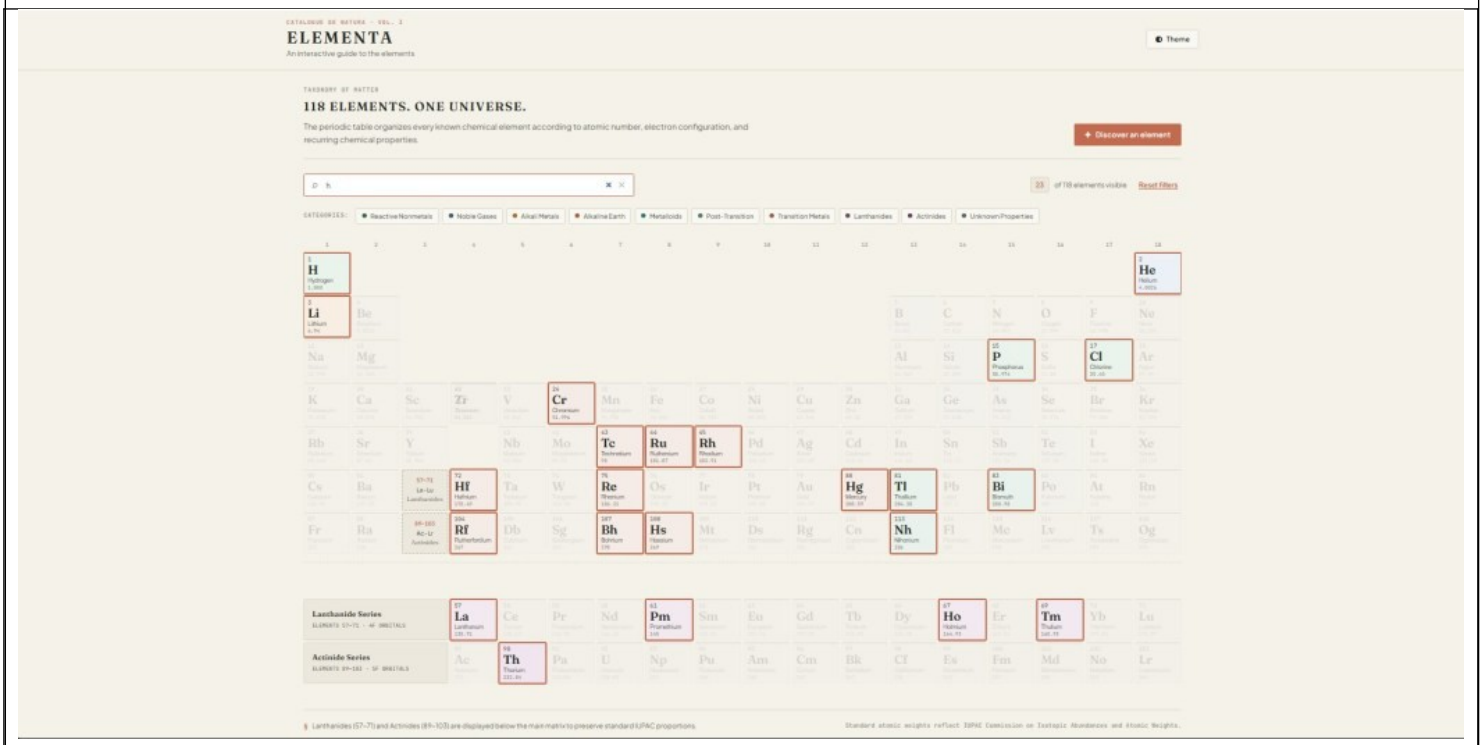
4. IMPLEMENTATION

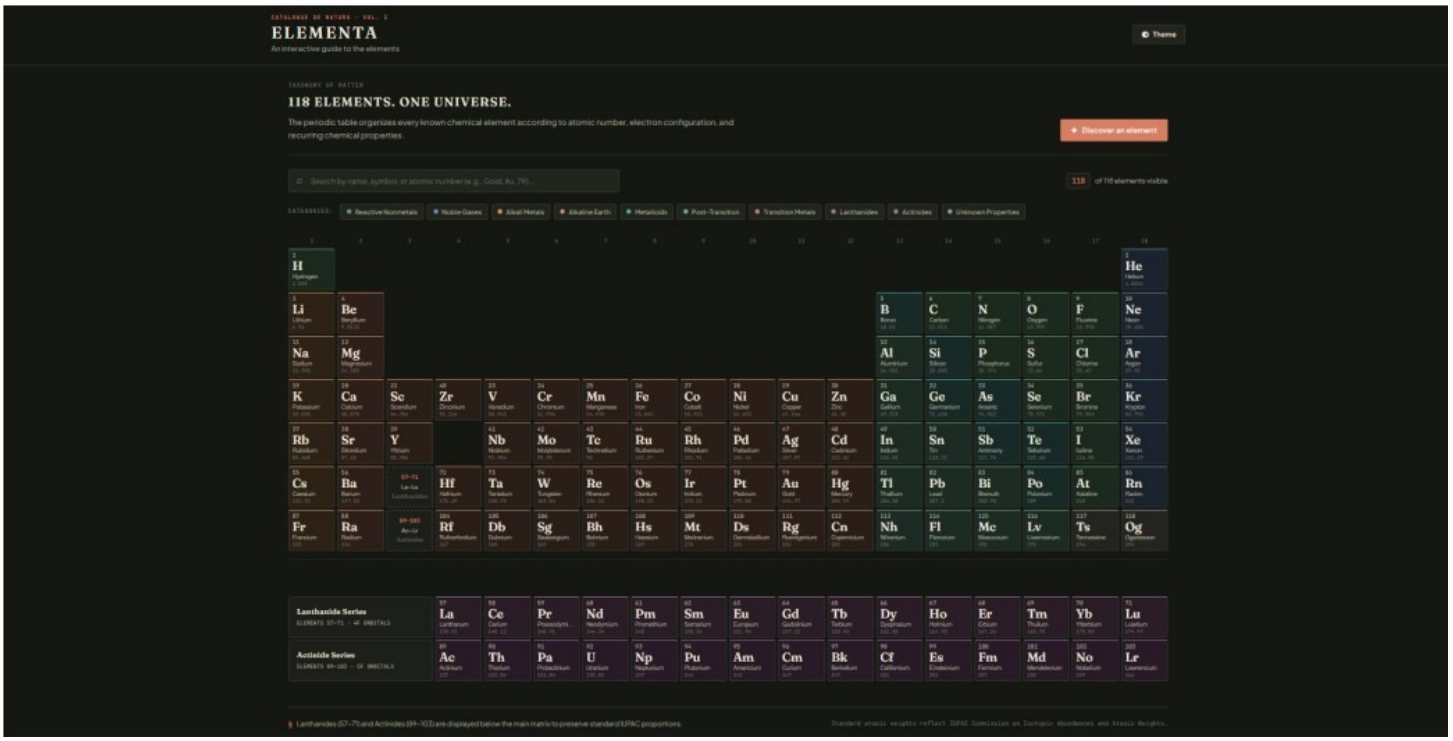
4.1 SCREENSHOTS

Field	Type	Example
number	Integer	79
symbol	String	"Au"
name	String	"Gold"
atomicMass	String	"196.967"
category	String	"transition-metal"
period	Integer	6
group	Integer	11
gridRow	Integer	6
gridCol	Integer	11
phase	String	"Solid"
electronConfig	String	"[Xe] 4f14 5d10 6s1"
electronegativity	String	"2.54"
meltingPoint	String	"1337.33 K"
boilingPoint	String	"3243 K"
density	String	"19.3 g/cm3"
discoveredBy	String	"Known since antiquity"
origin	String	"Latin aurum"
summary	String	"Used in electronics..."


CATALOGUE DE NATURA · VOL. I
ELEMENTA
An interactive guide to the elements
Theme
TAXONOMY OF MATTER
118 ELEMENTS. ONE UNIVERSE.
The periodic table organizes every known chemical element according to atomic number, electron configuration, and recurring chemical properties.
Discover an element
Search by name, symbol, or atomic number (e.g., Gold, Au, 79)...
118 of 118 elements visible
CATEGORIES:
Reactive Nonmetals
Noble Gases
Alkali Metals
Alkaline Earth
Metalloids
Post-Transition
Transition Metals
Lanthanides
Actinides
Unknown Properties
Lanthanide Series
ELEMENTS 57–71 · 4f ORBITALS
Actinide Series
ELEMENTS 89–103 · 5f ORBITALS
Lanthanides (57–71) and Actinides (89–103) are displayed below the main matrix to preserve standard IUPAC proportions.
Standard atomic weights reflect IUPAC Commission on Isotopic Abundances and Atomic Weights.


ELEMENT NO. 20
ALKALINE EARTH
20
Ca
40.078 u
Calcium
Latin 'calx' (lime)
Solid at STP · Period 4 · Group 2
OBSERVATION & USES
Essential structural component of teeth, bones, molluscan shells, and architectural concrete.
PHYSICAL & ATOMIC PROPERTIES
Electron Configuration
[Ar] 4s²
Electronegativity
1.0
Melting Point
1115 K (842 °C)
Boiling Point
1757 K (1484 °C)
Density
1.55 g/cm³
Discovery & History
Humphry Davy (1808)
Use ← and → arrow keys to navigate · Esc to close


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Lanthanides (57–71) and Actinides (89–103) are displayed below the main matrix to preserve standard IUPAC proportions.
Standard atomic weights reflect IUPAC Commission on Isotopic Abundances and Atomic Weights.

4.2 DATA DESIGN

There is no database. All data is held in a single JavaScript array const elements in script.js. Each element object has the following fields:

Field	Type	Example
number	Integer	79
symbol	String	"Au"
name	String	"Gold"
atomicMass	String	"196.967"
category	String	"transition-metal"
period	Integer	6
group	Integer	11
gridRow	Integer	6
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phase	String	"Solid"
electronConfig	String	"[Xe] 4f14 5d10 6s1"
electronegativity	String	"2.54"
meltingPoint	String	"1337.33 K"
boilingPoint	String	"3243 K"
density	String	"19.3 g/cm3"
discoveredBy	String	"Known since antiquity"
origin	String	"Latin aurum"
summary	String	"Used in electronics..."

4.3 CODE

DYNAMIC TABLE RENDERING (script.js)

```
function renderPeriodicTable() {
  const grid = document.getElementById('periodic-grid');
  grid.innerHTML = '';
  injectSeriesMarkers(grid);
  elements.forEach(el => {
    const tile = document.createElement('button');
    tile.className = `element-tile cat-${el.category}`;
    tile.style.gridRow = el.gridRow;
    tile.style.gridColumn = el.gridCol;
    tile.dataset.number = el.number;
    tile.dataset.symbol = el.symbol;
    tile.dataset.name = el.name.toLowerCase();
    tile.dataset.category = el.category;
    tile.setAttribute('aria-label', `${el.name}, element ${el.number}`);
    tile.innerHTML = `



      `,
    tile.addEventListener('click', () => openElementDetail(el.number));
    grid.appendChild(tile);
  });
}
```

UNIFIED SEARCH AND FILTER (script.js)

```
function applyFiltersAndSearch() {
  const query = searchInput.value.toLowerCase().trim();
  let visibleCount = 0;
  document.querySelectorAll('.element-tile').forEach(tile =>
    { const matchesQuery =
      !query ||
      tile.dataset.name.includes(query) ||
      tile.dataset.symbol.toLowerCase().includes(query) ||
      tile.dataset.number.includes(query);
    const matchesCategory = !
      activeCategoryFilter ||
      tile.dataset.category === activeCategoryFilter;
    if (matchesQuery && matchesCategory) {
      tile.classList.remove('dimmed');
      tile.classList.add('highlighted');
      visibleCount++;
    } else
      { tile.classList.add('dimmed')
      ;
      tile.classList.remove('highlighted');
    }
  });
  document.getElementById('visible-count').textContent =
visibleCount; }
```

ELEMENT DETAIL MODAL (script.js):

```
function openElementDetail(num) {
  const el = elements.find(item => item.number === num);
  if (!el) return;
  currentElementNumber = num;
  document.getElementById('modal-element-name').textContent = el.name;
  document.getElementById('modal-symbol').textContent = el.symbol;
  document.getElementById('modal-atomic-number').textContent = el.number;
  document.getElementById('modal-atomic-mass').textContent = el.atomicMass + ' u';
  document.getElementById('modal-electron-config').textContent = el.electronConfig;
  document.getElementById('modal-melting-point').textContent = el.meltingPoint;
  document.getElementById('modal-boiling-point').textContent = el.boilingPoint;
  document.getElementById('modal-density').textContent = el.density;
  document.getElementById('modal-electronegativity').textContent =
el.electronegativity;
  document.getElementById('modal-discovery').textContent = el.discoveredBy;
  document.getElementById('modal-latin-name').textContent = el.origin;
  document.getElementById('modal-description').textContent = el.summary;
  document.getElementById('modal-phase').textContent = el.phase;
  document.getElementById('modal-period').textContent = el.period;
  document.getElementById('modal-group').textContent = el.group;
  const modal = document.getElementById('detail-modal');
  modal.classList.remove('hidden');
  modal.setAttribute('aria-hidden', 'false');
  document.body.style.overflow = 'hidden';
}
```

THEME TOGGLE WITH LOCALSTORAGE (script.js)

```
function initTheme() {
  const saved = localStorage.getItem('elementa-theme') || 'light';
  document.documentElement.setAttribute('data-theme', saved);
}

document.getElementById('theme-toggle-
  btn') .addEventListener('click', () => {
    const current =
      document.documentElement .getAttribute(
      'data-theme');
    const next = current === 'light' ? 'dark' : 'light';
    document.documentElement.setAttribute('data-theme', next);
    localStorage.setItem('elementa-theme', next);
  });
```

CSS GRID LAYOUT (style.css)

```
  .periodic-grid
{ display: grid;
  grid-template-columns: repeat(18, minmax(64px, 1fr));
  grid-template-rows: repeat(10, minmax(72px, auto));
  gap: var(--gap-xs);
}

.table-viewport
  { overflow-x:
  auto;
  -webkit-overflow-scrolling: touch;
```

5. CONCLUSION

Elementa demonstrates that a complex, data-rich scientific reference tool can be built with pure HTML, CSS and JavaScript without any frameworks or build tools. The project provided practical experience in dynamic DOM generation, CSS Grid for complex layouts, real-time event-driven search and filtering, modal accessibility with keyboard navigation, and localStorage-based user preference persistence. The editorial design approach also demonstrated how thoughtful typography and colour choices can make technical content more readable and engaging. Future improvements could include a comparison mode to view two elements side by side, a quiz mode for students, more detailed spectral and isotope data, and a PWA offline mode.

6. REFERENCES

- MDN Web Docs (HTML5, CSS Grid, JavaScript DOM, localStorage), developer.mozilla.org
- IUPAC Commission on Isotopic Abundances and Atomic Weights, iupac.org
- Royal Society of Chemistry Periodic Table, rsc.org/periodic-table
- Google Fonts (Fraunces, Plus Jakarta Sans, JetBrains Mono), fonts.google.com
- Visual Studio Code Documentation, code.visualstudio.com
- WebAIM Accessibility Guidelines, webaim.org